Summary Report: Four Audits of One Judge Side by Side, v2, v3, v4 and v5 on the Same 31 Datasets and Ten Models

2026

1 Setup

Four audit pipelines were run on one cache of answers: 31 datasets (11 entity matching, 6 schema matching, 11 table QA, 3 finance), 100 stratified items each from the official splits (90 and 93 for two small sets), ten models, 3,083 answers per model, USD 161. Each version asks a different question of the deployed judge, whose rule is $R = 0.35\,\mathrm{F1} + 0.35\,e^{-2.5\,|\text{rel. err}|} + 0.15\,\mathrm{P} + 0.15\,\mathrm{R}$ ("v1"):

- **v2**: are v1's four weights the right ones? Every step-0.05 weighting of the four ingredients (1,771) is scored by how consistently the 31 datasets rank the models (mean Kendall τ to the Borda pool), with correctness on 31,380 synthetic answers of known quality checked beside it, nested-bootstrap selection, a plateau, leave-one-out transfer and sampled designs.
- **v3**: given per-dataset scores, how should they be pooled? Eight aggregation views from the literature (mean, baseline-normalised mean, mean z, win rate, Borda, Kemeny, Bradley–Terry, Rasch) and all 255 equal-weight subsets plus 200 sampled mixes, scored by stability under item resampling, transitivity and agreement with the views' consensus; split-half honesty; parameters with standard errors.
- **v4**: does a richer per-answer rule help? Nine lexical components, three sampled designs of 100 weightings plus baselines, the same agreement objective and honesty battery as v2.
- **v5**: what does the external quality signal say? A 100-metric library reduced to three by anchor-driven selection, an exhaustive grid over their weights and a hedge penalty scored by correctness on the anchors, a top-100 ensemble, leave-one-dataset-out re-selection and utility perturbation.

All four share the pools, the answer cache, the scoring primitives, the coverage rule and the leaderboard machinery (Bradley–Terry headline with Borda, Kemeny, Copeland and mean-z checks; 1,000 item bootstraps; Friedman, Nemenyi and Holm across datasets; per-item paired tests; Kendall's W; post-stratified official-split estimates). Table 6 summarises the design differences. Each version's full Setup, Results, figures, tables and data are in the sibling packages `v2/`, `v3/`, `v4/`, `v5/`.

2 Results side by side

2.1 Which rule scores answers best

On the one quantity all per-answer rules can be compared on, correctness against the same 31,380 anchors, the order is: v5 ensemble ρ 0.904 > v4 best-ρ 0.884 $\approx$ v2 best-ρ 0.883 > v1 as deployed 0.878 > token F1 alone 0.859 > exact match 0.828 > numeric tolerance alone (v4's best by agreement) 0.796 (Figure 1, Table 5). Every rule separates correct from wrong answers with AUC $\geq$ 0.99; the differences are in graded credit for near misses, unit variants and partial answers. The deployed v1 is within 0.026 of the best rule found by any version.

Table 1: What each version does.

	v2	v3	v4	v5
per-answer rule	4 metrics (F1, decay, P, R), weights searched	dataset's own metric	9 metrics, weights searched	3 of 100 metrics, weights searched, top-100 ensemble, hedge gate
what is searched	4 weights, exhaustive (1,771)	mix of 8 aggregation views (455 candidates)	9 weights, 3 sampled designs (305)	3 weights + lambda, exhaustive (6,630)
objective	datasets agree (tau to Borda pool)	stability + transitivity + consensus	datasets agree (tau to Borda pool)	anchor rho, AUC, JS consistency, native tau, wrong -> 0
checked against	the ranked models (+ anchors beside)	bootstrap resamples	the ranked models (+ anchors beside)	31,380 synthetic answers of known quality
honesty test	nested item bootstrap, plateau, LODO/LOFO	split-half vs pure BT / uniform	nested item bootstrap, plateau, LODO/LOFO	LODO re-selection, utility perturbation, probes
headline leaderboard rule	v1 weights (deployed)	native metric, Bradley-Terry	v1 rubric (comparability)	calibrated ensemble
main risk	circular: agreement is not correctness	views differ in kind; no per-answer rule	circular; 9-D space not enumerable	hand-set utility scale; synthetic anchors

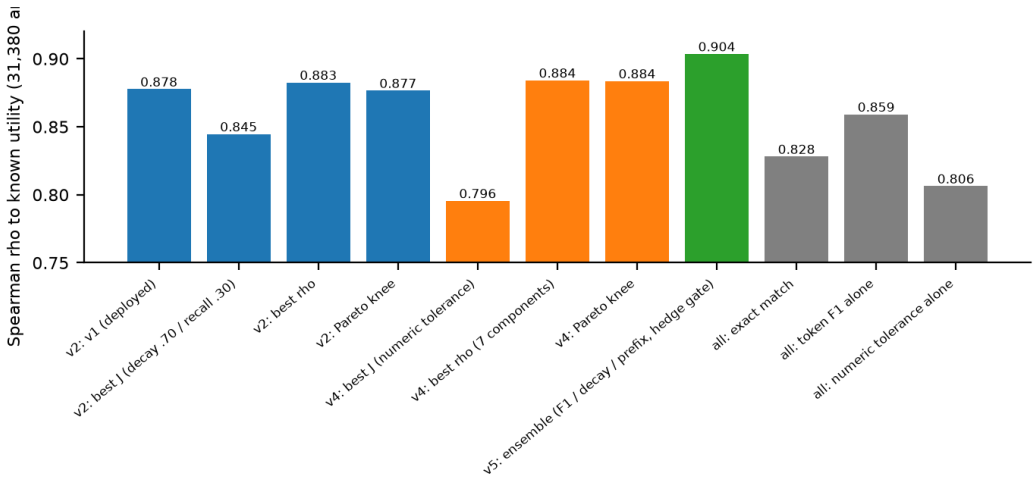


Figure 1: Correctness of each version's scoring rule on the same 31,380 anchors (v3 has no per-answer rule).

Table 2: Anchor correctness of each version's rule on the same 31,380 synthetic answers.

version	rule	rho	auc	wrong mean
v2	v1 (deployed)	0.878	0.994	0.034
v2	best J (decay .70 / recall .30)	0.845	0.994	0.037
v2	best rho	0.883	0.994	0.034
v2	Pareto knee	0.877	0.994	0.036
v4	best J (numeric tolerance)	0.796	0.989	0.022
v4	best rho (7 components)	0.884	0.994	0.053
v4	Pareto knee	0.884	0.994	0.039
v5	ensemble (F1 / decay / prefix, hedge gate)	0.904	0.996	0.032
all	exact match	0.828	0.989	0.011
all	token F1 alone	0.859	0.965	0.032
all	numeric tolerance alone	0.806	0.919	0.016

2.2 Which search finds something real

Under the agreement objective (v2, v4) the in-sample winner loses about 0.05 when held out and ends within noise of v1 (-0.008 for v2, -0.002 for v4); the plateau covers almost every candidate in both. v3's split-half test gives the same picture for the choice of pooling rule: the selected mix beats pure Bradley–Terry by +0.024 with an interval through zero, and the uniform mix does as well. v5 is the only version whose selection survives its honesty test: the same three metrics are chosen in 90 % of leave-one-dataset-out folds, held-out ρ moves by 0.001, and the result is +0.080 above the native typed metric on the same folds (Figure 2, Table 8). The reason is the objective, not the search: agreement among the ranked models is flat over the weight space, correctness against an external signal is not.

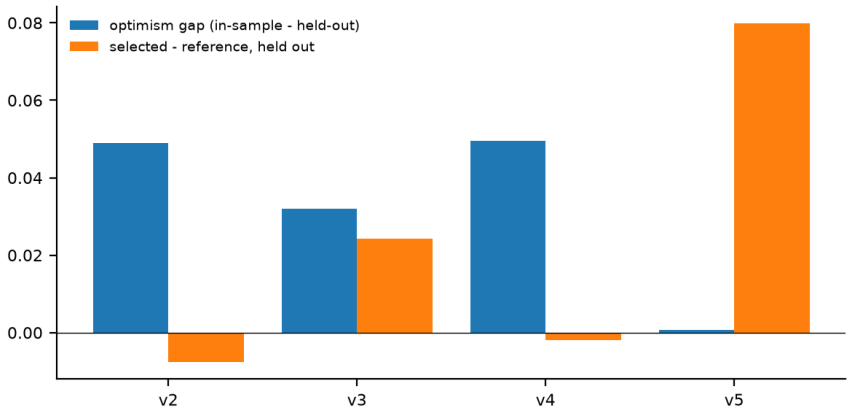


Figure 2: Honest-selection outcome per version: how much the in-sample winner loses when held out, and whether it still beats the version's reference rule.

Table 3: Honest-selection outcome per version.

version	test	optimism gap	selected - reference (held out)	reference	distinct selections	plateau
v2	nested item bootstrap (B = 1,000)	0.049	-0.008	v1 weights	608	1765 of 1,771
v3	split-half (20 splits)	0.032	0.024	pure Bradley-Terry	10	–
v4	nested item bootstrap (B = 1,000)	0.050	-0.002	v1 rubric	204	305 of 305
v5	leave-one-dataset-out re-selection + utility perturbation	0.001	0.080	native typed metric	same set in 90% of folds	top-100 ensemble

2.3 Transfer and designs

Re-fitted weights predict a held-out dataset no better than v1 in v2 or v4 (Figure 3). Sampled designs find the optimum whenever it exists (regret ≤ 0.005 in v2, ≤ 0.0015 in v5); in v4's nine-dimensional space the uniform lattice has the best coverage and the best found value, and the three designs agree to 0.01 (Figure 4).

2.4 The leaderboard every version produces

The same answers give the same ends under every rule and every version: Llama 3.3 70B last, gpt-oss-120b ninth, and Claude Opus 5, GPT-5.5, Gemini 3.8 Flash as the top three, with Friedman $p < 0.001$ and Holm-significant separation of the top group from the bottom two (Table 10). What

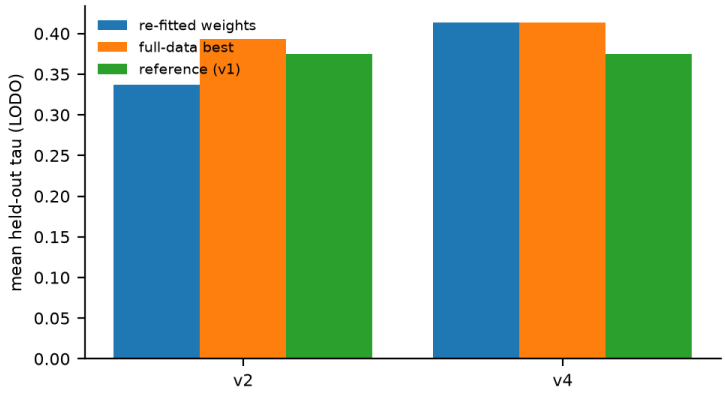


Figure 3: Leave-one-dataset-out transfer for the two versions that search per-answer weights under the agreement objective.

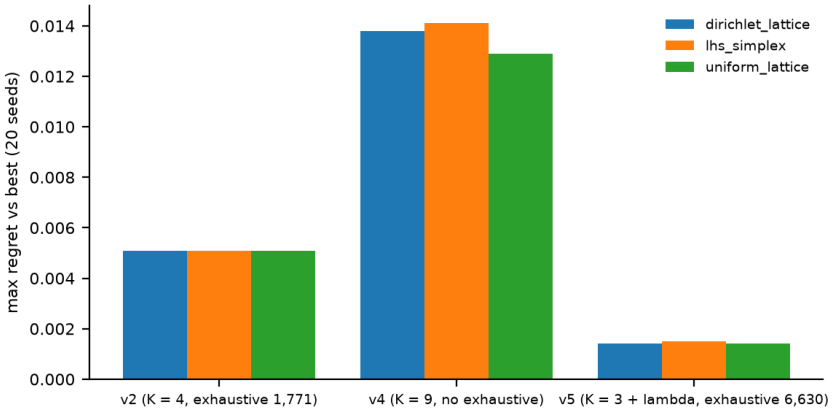


Figure 4: Sampled-design regret per version: with few weights every design finds the optimum; with nine the lattice design is marginally better.

changes is the order inside the top three (Figure 5): exact match and v4's numeric-tolerance rule put Opus first; the graded rules (v1, v2 best, v4 best-ρ, v5 ensemble) put Gemini 3.8 Flash first by Bradley–Terry and GPT-5.5 or Opus first by Borda; v3's native-metric pooling puts Gemini Flash first by Bradley–Terry and GPT-5.5 by Borda. Kendall's W among datasets is 0.25–0.28 everywhere: the datasets disagree about the middle of the table regardless of rule.

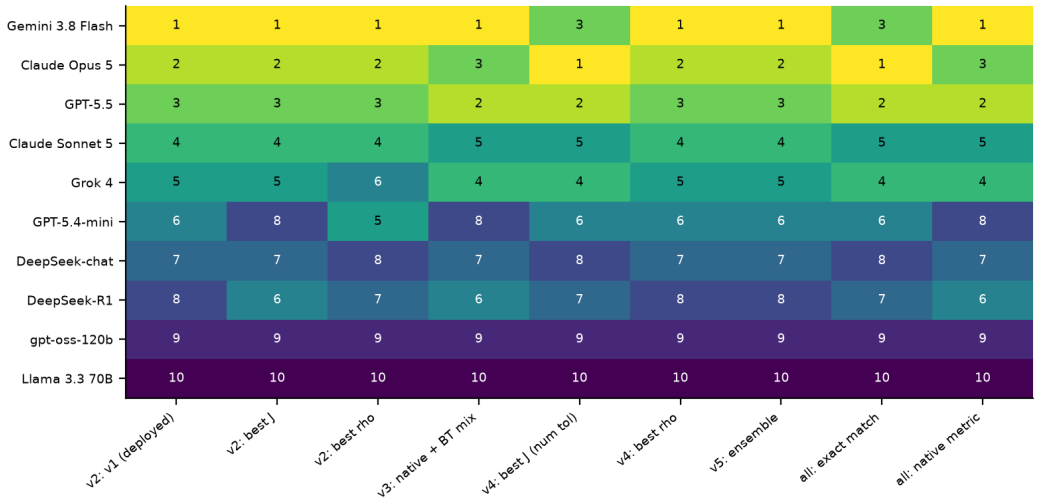


Figure 5: Bradley–Terry rank of every model under each version's headline rule(s) and the two common baselines.

Table 4: Leaderboard significance per headline rule.

version	rule	Kendall W	Friedman p	Holm-significant pairs	first (BT)	first (Borda)	last
v2	v1 (deployed)	0.264	0.000	13	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B
v2	best J	0.264	0.000	12	Gemini 3.8 Flash	Claude Opus 5	Llama 3.3 70B
v2	best rho	0.256	0.000	12	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B
v3	native + BT mix	0.274	0.000	13	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B
v4	best J (num tol)	0.276	0.000	13	Claude Opus 5	Claude Opus 5	Llama 3.3 70B
v4	best rho	0.272	0.000	12	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B
v5	ensemble	0.259	0.000	11	Gemini 3.8 Flash	Claude Opus 5	Llama 3.3 70B
all	exact match	0.279	0.000	12	Claude Opus 5	Claude Opus 5	Llama 3.3 70B
all	native metric	0.283	0.000	13	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B

3 What is better, per version

v2 (exhaustive grid over v1's weights). Best at answering the question it was built for: it proves, not samples, that no weighting of v1's ingredients is distinguishable from v1 on this suite (plateau 1,765 of 1,771; out-of-bag difference −0.008 with an interval through zero). It is the right headline for "the deployed weights are fine". It cannot say the weights are *right*, because agreement is not correctness.

v3 (aggregation views). Best at the pooling question and the only version that reports fitted parameters with standard errors (Bradley–Terry, Rasch, Kemeny cost). Its finding is that mixing views buys little over the uniform mix and that pure Bradley–Terry is the least reproducible single rule on 31 datasets × 10 models. It has no per-answer rule, so it cannot audit the judge's formula.

v4 (nine components). Best as the negative control: adding five components and sampling 305 candidates in a 3-million-point space finds that a single typed metric (numeric tolerance)

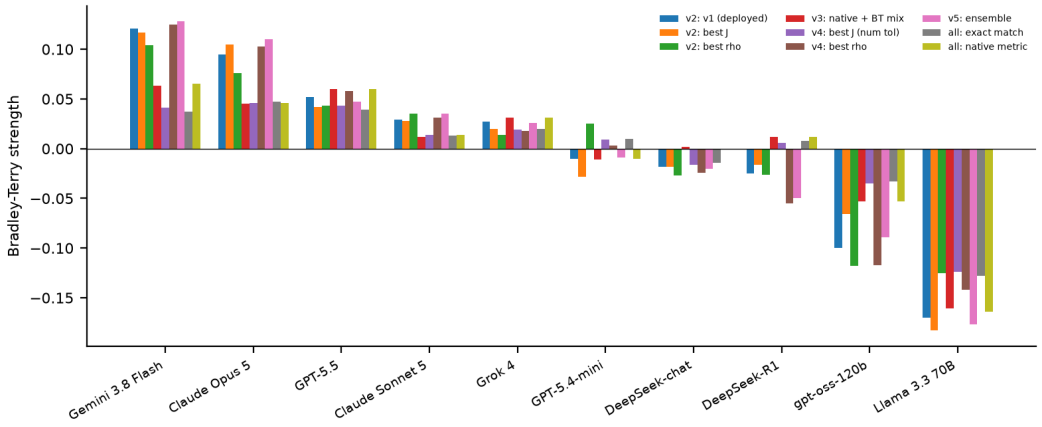


Figure 6: Bradley–Terry strengths per model under each version's headline rule(s).

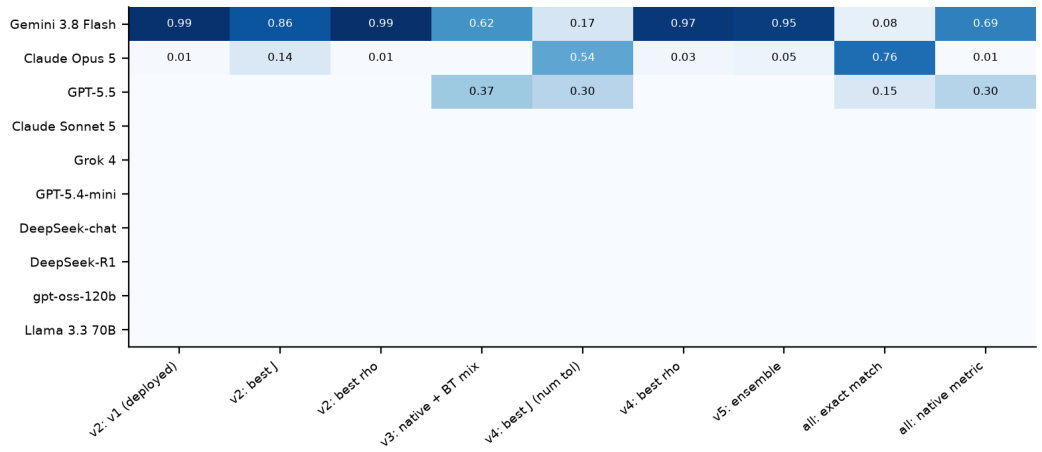


Figure 7: Bootstrap probability of first place under each headline rule.

maximises agreement, and that this metric is the worst on correctness. It shows most clearly that the agreement objective rewards coarseness. Its search cannot be exhaustive, so its optimum is only "best found".

v5 (anchor-calibrated ensemble). Best scoring rule by the external signal (ρ 0.904, AUC 0.996) and the only selection that survives leave-one-dataset-out unchanged. It is the version to deploy if the goal is to score answers well; its weakness is that the signal is synthetic (a hand-set utility scale) until human labels exist, and its gain over v1 (+0.026 in ρ) is real but small.

Taken together. v2 and v4 establish that the weight question has no answer on this suite; v3 establishes that the pooling question has no strong answer either; v5 establishes that the correctness question does, and that the deployed rule is close to its answer. The paper's claim is therefore: keep v1 (or the v5 ensemble), report the leaderboard under several rules with the ends and the significant pairs, and spend the next effort on human labels rather than on more weight search.

A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

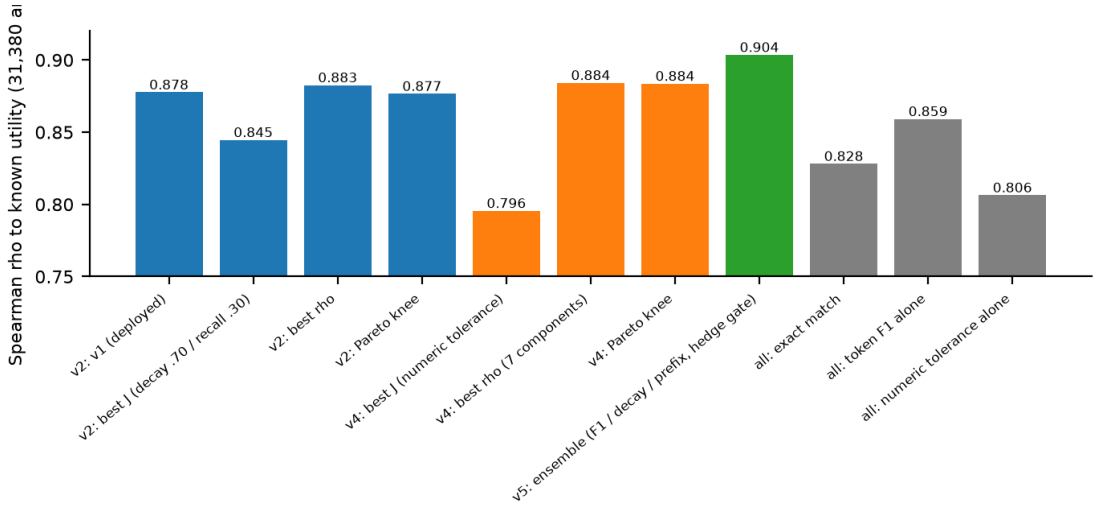


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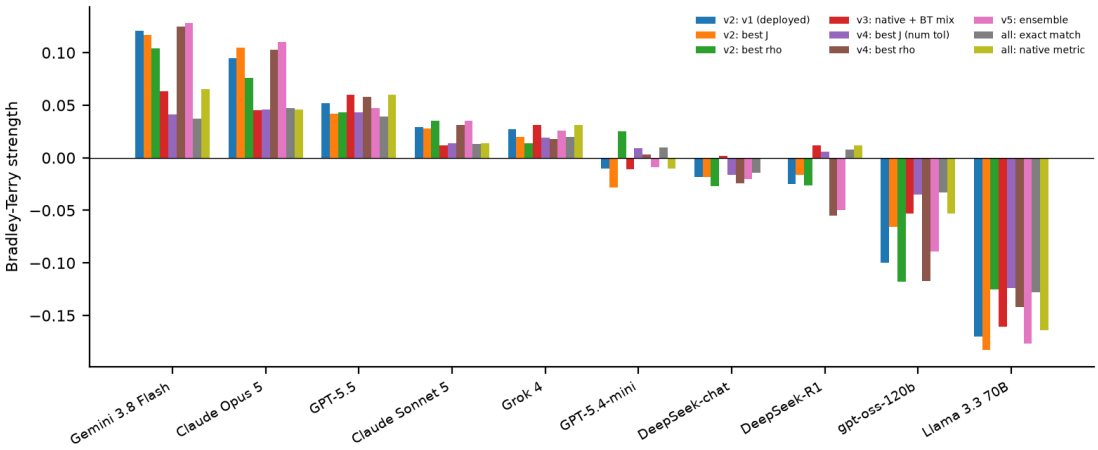


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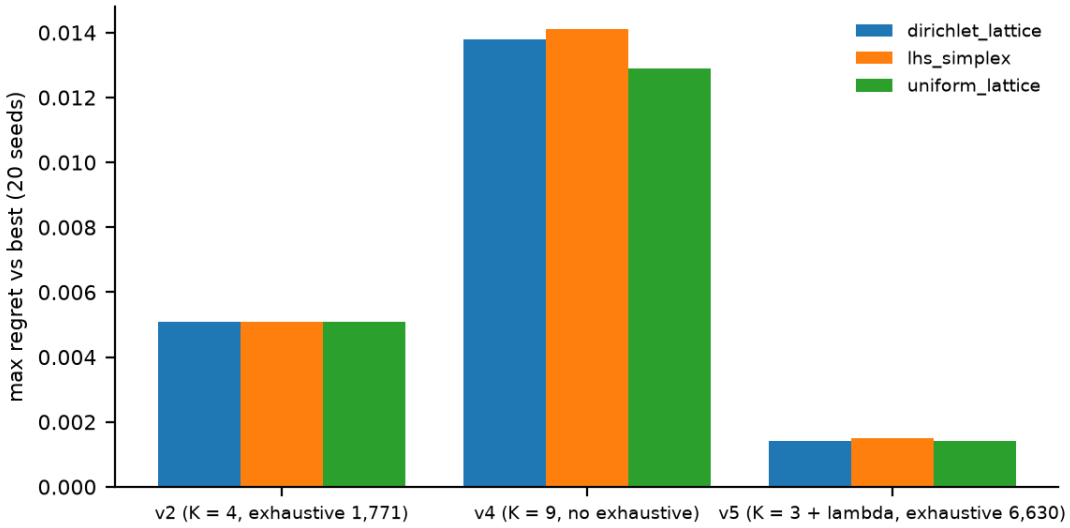


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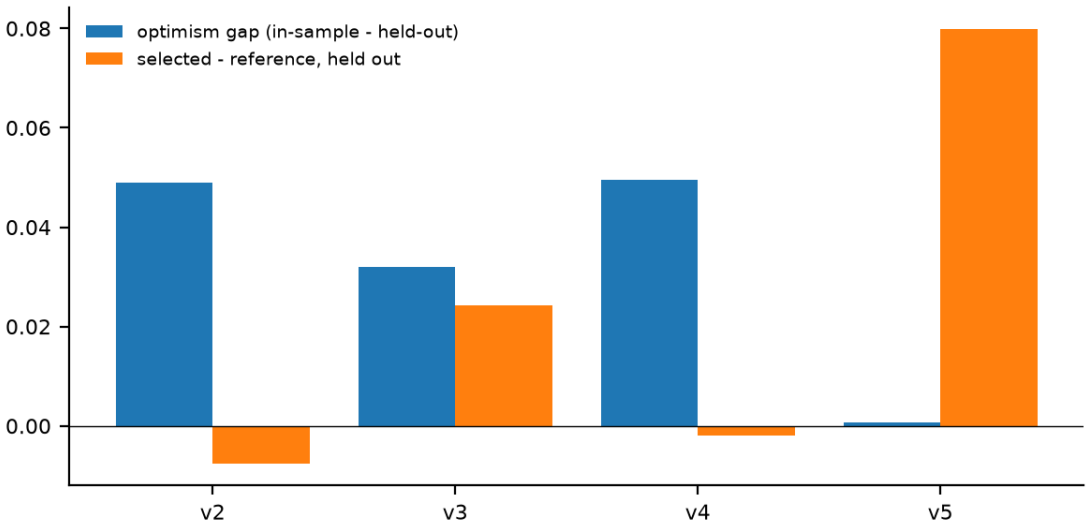


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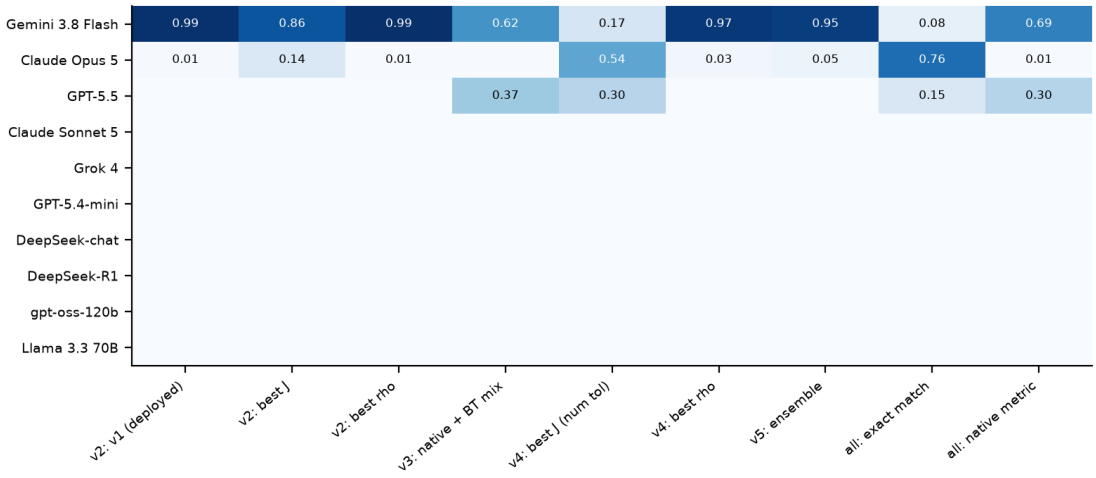


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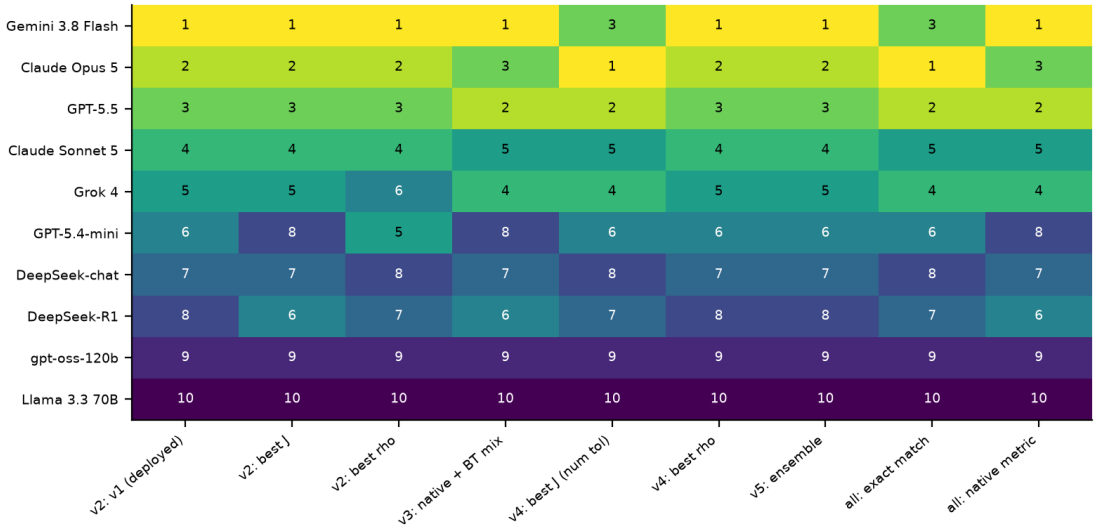


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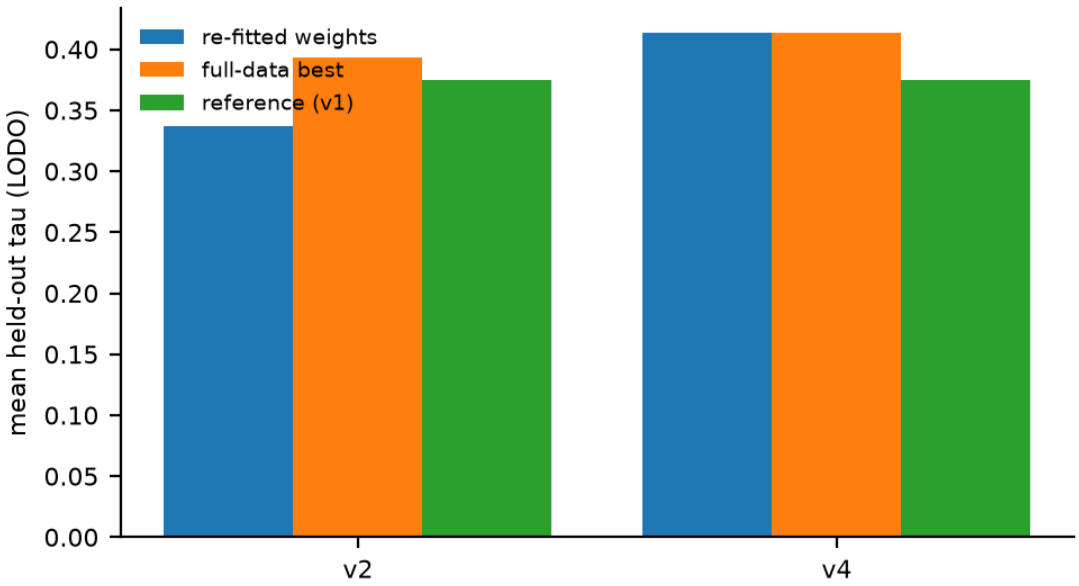


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B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

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Table 6: What each version does.

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main risk	circular: agreement is not correctness	views differ in kind; no per-answer rule	circular; 9-D space not enumerable	hand-set utility scale; synthetic anchors

Table 7: Sampled-design regret per version.

version	design	regret mean	regret max
v2 (K = 4, exhaustive 1,771)	dirichlet_lattice	0.0038	0.0051
v2 (K = 4, exhaustive 1,771)	lhs_simplex	0.0034	0.0051
v2 (K = 4, exhaustive 1,771)	uniform_lattice	0.0029	0.0051
v4 (K = 9, no exhaustive)	dirichlet_lattice	0.0100	0.0138
v4 (K = 9, no exhaustive)	lhs_simplex	0.0107	0.0141
v4 (K = 9, no exhaustive)	uniform_lattice	0.0077	0.0129
v5 (K = 3 + lambda, exhaustive 6,630)	dirichlet_lattice	0.0007	0.0014
v5 (K = 3 + lambda, exhaustive 6,630)	lhs_simplex	0.0008	0.0015
v5 (K = 3 + lambda, exhaustive 6,630)	uniform_lattice	0.0007	0.0014

Table 8: Honest-selection outcome per version.

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v5	leave-one-dataset-out re-selection + utility perturbation	0.001	0.080	native typed metric	same set in 90% of folds	top-100 ensemble

Table 9: Bradley–Terry rank per model under each headline rule.

model	v2: v1 (deployed)	v2: best J	v2: best rho	v3: native + BT mix	v4: best J (num tol)	v4: best rho	v5: ensemble	all: exact match	all: native metric
Gemini 3.8 Flash	1	1	1	1	3	1	1	3	1
Claude Opus 5	2	2	2	3	1	2	2	1	3
GPT-5.5	3	3	3	2	2	3	3	2	2
Claude Sonnet 5	4	4	4	5	5	4	4	5	5
Grok 4	5	5	6	4	4	5	5	4	4
GPT-5.4-mini	6	8	5	8	6	6	6	6	8
DeepSeek-chat	7	7	8	7	8	7	7	8	7
DeepSeek-R1	8	6	7	6	7	8	8	7	6
gpt-oss-120b	9	9	9	9	9	9	9	9	9
Llama 3.3 70B	10	10	10	10	10	10	10	10	10

Table 10: Leaderboard significance per headline rule.

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all	native metric	0.283	0.000	13	Gemini 3.8 Flash	GPT-5.5	Llama 3.3 70B

Table 11: Leave-one-dataset-out transfer (mean held-out τ).

version	re-fitted weights	full-data best	reference (v1)
v2	0.337	0.393	0.374
v4	0.413	0.413	0.374